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CLASSROOM RESEARCH IN GEOGRAPHY

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CHAPTER 1: INTRODUCTION AND BACKGROUND

1.1 Background to the Problem

Geography is an important subject in schools around the world. It teaches learners how to think about space, the environment, and how people interact with the world around them. One key skill in Geography is mapwork — the ability to read, interpret, and use different types of maps. Across the world, scholars such as Bednarz and Schee (2007) have noted that map skills are becoming more important in modern education, especially as technology like GIS becomes part of everyday life. Despite this, many learners still struggle to understand and use maps correctly.

In South Africa, the Curriculum and Assessment Policy Statement (CAPS) for Geography, Grades 10 to 12, requires all learners to learn mapwork as a compulsory part of the subject (Department of Basic Education, 2011). Learners are expected to read contour lines, calculate scale and distance, identify grid references and bearings, and interpret aerial photographs. These skills are tested every year in the National Senior Certificate (NSC) examinations, where mapwork makes up a significant part of the Geography paper. Despite this, mapwork is one of the sections where learners perform the worst nationally.

In Limpopo Province, the problem is made worse by challenges in schools such as limited resources, large classes, and a lack of learning materials (Du Plessis, 2014). In the Pietersburg Circuit, which falls under the Capricorn South District, these problems combine with the already difficult nature of mapwork to create a serious barrier for learners. Olaniran et al. (2019) found that many Geography teachers in South Africa struggle to teach mapwork well, which means learners often go into tests and examinations without a solid understanding of the content. This study will focus on this problem in the Pietersburg Circuit.

1.2 Problem Statement

Grade 10 Geography learners in the Pietersburg Circuit regularly perform poorly in mapwork. This is confirmed by national NSC diagnostic reports (Department of Basic Education, 2011), by the experiences of Geography teachers in the circuit who say learners are confused by contour lines, scale, and direction, and by the research of

Wilmot and Dube (2016), who found that many South African Geography learners lack the spatial thinking skills needed for mapwork. The difficulties learners face are serious. They struggle to picture a three-dimensional landscape from a flat map, they cannot convert a map scale into a real-world distance, and they misread examination instructions and apply the wrong method.

Despite how widespread this problem is, very little research has been done specifically in the Pietersburg Circuit. National and provincial studies have identified mapwork as a problem area, but there is no in-depth local study that looks at exactly why learners in this circuit struggle, and what can be done about it. This study will fill that gap by looking closely at the cognitive and teaching challenges that affect mapwork learning among Grade 10 Geography learners in selected schools in the Pietersburg Circuit.

1.3 Aim and Objectives

1.3.1 Research Aim

The aim of this study is to **explore and understand the cognitive and teaching challenges** that Grade 10 Geography learners face when learning mapwork in the Pietersburg Circuit, Capricorn South District, Limpopo Province, South Africa, and to identify how teaching support can be improved.

1.3.2 Research Objectives

1. To identify the specific thinking challenges learners face when trying to picture three-dimensional landforms from two-dimensional contour lines on a map.
2. To find out how weaknesses in mathematical skills affect a learner's ability to calculate and understand map scale and distance.
3. To examine how the inability to read and understand examination instructions affects how learners perform in mapwork tasks.
4. To explore the challenges of using GIS and digital tools in schools in the Pietersburg Circuit that have limited resources.

1.4 Research Questions

Main Research Question: What cognitive and teaching challenges do Grade 10 Geography learners face when learning mapwork in the Pietersburg Circuit, Capricorn South District, Limpopo Province?

Sub-questions:

- Which mapwork topics (contour lines, scale, bearings, and cross-sections) do Grade 10 learners in Pietersburg Circuit find hardest, and why?
- How do weak mathematical skills make it harder for learners to calculate scale and measure distances on maps?
- How does misreading examination instructions affect learner performance in mapwork tasks?
- What effect do a lack of resources and limited teacher knowledge have on how mapwork is taught in the Pietersburg Circuit?
- How can Vygotsky's ZPD and scaffolding be used to improve how mapwork is taught in Grade 10 Geography?

1.5 Rationale and Significance of the Study

This study is important for two reasons: it adds to knowledge about Geography education, and it has practical value for teachers and learners.

From a knowledge perspective, this study will contribute to what is already known about spatial thinking and Geography education in South Africa. By using Vygotsky's (1978) sociocultural theory as a guide, the study will help explain why learners struggle with mapwork and how better teaching support could make a difference.

From a practical perspective, the findings will give Geography teachers in Pietersburg Circuit a clearer picture of which mapwork topics learners find most difficult and why. This can directly help teachers plan better lessons, choose the right teaching strategies, and develop additional support materials. The recommendations from this study will be realistic — taking into account the limited resources available in most schools in the circuit.

The main beneficiaries of this study will be Grade 10 Geography learners and their teachers in the Pietersburg Circuit. Other beneficiaries will include curriculum advisors in the Capricorn South District, the Limpopo Department of Education, and researchers working on Geography education. Because very little research has been done at circuit level in Limpopo, this study will fill an important gap and add a local South African perspective to the wider conversation about map literacy in schools.

1.6 Definition of Key Terms

Mapwork: The reading, interpretation, and use of maps such as topographic maps, orthophoto maps, and aerial photographs to understand things like scale, direction, contour lines, and geographical features. In South Africa's CAPS curriculum, mapwork is a compulsory part of Grade 10 to 12 Geography (Department of Basic Education, 2011).

Contour Lines: Lines on a topographic map that connect points that are at the same height above sea level. By looking at how the lines are spaced and shaped, a reader can figure out what the landscape looks like in three dimensions.

Map Scale: A way of showing the relationship between a distance on a map and the actual distance on the ground. For example, a scale of 1:50 000 means that 1 cm on the map equals 50 000 cm (500 m) on the ground.

Spatial Cognition: The mental ability to understand, picture, and reason about space. In Geography, this includes being able to mentally 'build' a three-dimensional picture of a landscape from a flat, two-dimensional map (Wilmot & Dube, 2016).

Zone of Proximal Development (ZPD): A concept from Vygotsky (1978) that describes the gap between what a learner can do on their own and what they can do with help from a more knowledgeable person.

Scaffolding: The temporary help and support a teacher or peer provides to help a learner complete a task they could not yet do alone. As the learner improves, the support is gradually reduced (Verenikina, 2003).

Pietersburg Circuit: An educational circuit within the Capricorn South District of the Limpopo Department of Education, serving secondary schools in and around the Polokwane area of Limpopo Province.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter looks at what researchers and scholars have already found about learners' challenges with mapwork, the ways mapwork is taught in schools, and the broader factors that affect how well learners perform in mapwork. The review is organised around four main problem areas identified in this study:

- the challenge of working with map scale
- the difficulty of picturing a three-dimensional landscape from a two-dimensional map;
- the role of being able to read and understand examination instructions; and
- the impact of limited resources and teaching support in Limpopo Province.

The chapter ends by introducing Vygotsky's Zone of Proximal Development as the theoretical framework for this study. The review shows that, while research on mapwork challenges exists at the national and international level, very little has been done at the circuit level, particularly in the Pietersburg Circuit.

2.2 Learner Challenges in Mapwork: Four Key Problem Areas

2.2.1 Working with Map Scale: The Mathematical-Spatial Problem

One of the most common challenges learners face in Geography is understanding and using map scale. Lane (2015) found that reading a ratio scale (such as 1:50 000, which is used on South African topographic maps) requires a level of thinking that many secondary school learners have not yet fully developed. The problem is not just about doing the maths correctly. Learners need to understand what the numbers actually mean in the real world: that 1 cm on the map equals a specific, measurable distance on the ground. Learners who treat scale as a purely mathematical exercise, without connecting it to physical space, often make mistakes even when they apply the right formula.

This problem is made worse in South Africa because many learners already struggle with basic mathematical concepts like ratio, proportion, and unit conversion (Du Plessis, 2014). For Grade 10 learners in Limpopo in particular, mapwork scale tasks require them to use both mathematical and spatial thinking at the same time, which puts a heavy load on their thinking. Wilmot and Dube (2016) found that even student

Geography teachers (who have already completed secondary Geography) continue to make errors in map scale tasks. This suggests that the problem is not being properly addressed through current teaching methods.

2.2.2 Reading Contour Lines: The 3D Visualisation Problem

Contour lines are one of the most difficult mapwork topics for learners to understand. They require a learner to take a flat, two-dimensional pattern of lines and mentally convert it into a three-dimensional picture of a landscape — seeing hills, valleys, steep slopes, and gentle slopes from lines printed on a page. This ability is known as graphicacy, and researchers treat it as a separate skill (similar to numeracy or reading) that has to be developed through practice and good teaching.

Wilmot and Dube (2016) showed that many South African Geography learners struggle with this skill. They found that learners often cannot picture what a landscape looks like just by looking at its contour lines, and they cannot determine whether a slope is steep or gentle from how close together the lines are. This means learners regularly make errors when drawing cross-sections, identifying landforms, and calculating vertical intervals. These errors are systematic (they come back in test after test) and they are difficult to fix through practice alone if the underlying spatial thinking has not been developed.

Bednarz and Schee (2007) suggest that GIS technology could help learners develop this spatial thinking by letting them see the relationship between the flat map and the actual terrain. However, as discussed in Section 2.2.4, most schools in the Pietersburg Circuit do not have the technology or equipment needed for this approach, so teachers must find other ways to help learners picture the three-dimensional world from a flat map.

2.2.3 Reading Examination Instructions: The Language Problem

A third challenge that is often overlooked in the literature is learners' difficulty with understanding what mapwork examination questions are actually asking them to do. Mapwork questions use very specific language. For example, the instruction 'determine' usually means the learner must calculate an answer, 'describe' asks for an observation, 'explain' asks for a reason, and 'calculate' asks the learner to use a specific formula. If a learner confuses these instructions; for example, by writing a

description when the question asks for a calculation; they will get no marks, even if their geographical understanding is sound.

Olaniran et al. (2019) found that Geography teachers in South African schools often point to this kind of confusion as a major reason why learners lose marks in mapwork. In Limpopo, where many learners are taught in a language that is not their first language, this challenge becomes even greater. The result is that a learner's mapwork performance does not always reflect what they actually know — it is also shaped by how well they can read and respond to the specific language used in examinations.

2.2.4 Limited Resources and Teaching Support: The Limpopo Context

The fourth challenge is about the conditions in which mapwork is taught in the Pietersburg Circuit. Du Plessis (2014) has documented serious resource shortages in Limpopo Province, including a lack of up-to-date textbooks, limited access to topographic and orthophoto maps, and a shortage of basic tools like protractors, rulers, and string; all of which are needed for mapwork activities. When learners do not have the right tools to practise skills like measuring distance on a map or calculating a bearing, it is very difficult for them to develop the level of accuracy that examinations require.

Olaniran et al. (2019) also found something important: some Geography teachers in South African schools avoid spending too much time on mapwork because they are not confident in their own knowledge of the topic. They focus on areas they find easier to teach, which means learners in those classes get less practice with mapwork than they need. This is a serious concern, because it means that the challenges learners face are not only about their own abilities; they are also shaped by the support, or lack of support, they receive from their teachers.

2.3 How Mapwork Is Currently Taught

Research suggests that there are better and worse ways to teach mapwork. Jo and Bednarz (2014) argue that spatial thinking is not something a person is simply born with — it can be taught and developed through the right kinds of activities. Approaches that work well include building physical models of landscapes using clay, tracing cross-sections on clear overlays, and comparing a topographic map with an aerial photograph of the same area. These hands-on activities help learners connect the flat map to the real landscape.

However, Olaniran et al. (2019) found that in many South African schools, mapwork is taught mainly through textbooks and examination practice — without these hands-on activities. Learners are shown how to perform steps and procedures but are not always helped to understand why those steps work or what the results mean in the real world. This gap between how mapwork is taught and what it actually requires learners to understand helps explain why poor performance persists year after year.

2.4 Factors That Affect Mapwork Performance

Many different factors contribute to how well or poorly learners perform in mapwork. At the thinking level, Wilmot and Dube (2016) identify weak spatial cognition as the biggest barrier. At the language level, the combination of difficult academic vocabulary, technical mapwork terms, and learning in a second or third language creates extra challenges for many South African learners. At the teaching level, Olaniran et al. (2019) point to teacher avoidance and too little time spent on mapwork instruction. And at the school level, Du Plessis (2014) shows that limited resources and poor infrastructure in Limpopo make it harder for teachers to deliver good mapwork lessons.

What all of these factors have in common is that they point to the importance of good, supportive teaching. When learners receive clear, step-by-step guidance from a knowledgeable teacher, they are much more likely to develop the understanding they need. This is exactly what Vygotsky's theory of the Zone of Proximal Development is about — which is why it has been chosen as the theoretical framework for this study.

2.5 Theoretical Framework: Vygotsky's Zone of Proximal Development

This study will be guided by the sociocultural theory of Lev Vygotsky (1978), and in particular by his concept of the Zone of Proximal Development (ZPD). Vygotsky believed that learning happens through interaction with other people, particularly with teachers or more knowledgeable peers who provide the support a learner needs to move forward. The ZPD is the space between what a learner can do on their own and what they can do with guidance. According to Vygotsky, the best teaching happens within this space — where the learner is being stretched but is supported enough to succeed.

Shabani, Khatib, and Ebadi (2010) explain that the main tool for teaching within the ZPD is scaffolding — the temporary, step-by-step support a teacher provides to help a learner complete a task they could not yet do alone. As the learner becomes more

competent, the support is gradually reduced until the learner can work independently. Verenikina (2003) points out that genuine scaffolding requires the teacher to carefully assess where the learner is struggling, and then provide exactly the kind of help that will push the learner forward — not just give them the answer.

In mapwork, this framework is very useful. The research reviewed in this chapter consistently shows that learners who do not receive proper guidance from a knowledgeable teacher; whether because the teacher is not confident in the topic, because class sizes are too large, or because resources are insufficient; are left to work within the ZPD without any support. As a result, they stay stuck at the level they are already at and cannot make progress. Mncube, Mncube, and Mthembu (2018) found this same pattern in South African classrooms, showing that a lack of proper teaching support is a major reason why learners underperform in challenging subjects like Geography.

2.6 Conclusion

The literature reviewed in this chapter shows that mapwork challenges in Grade 10 Geography come from several different directions: cognitive difficulties, language barriers, weak teaching support, and limited school resources. While research at the national and international level has given us a good general understanding of these challenges, no study has looked specifically at learners in the Pietersburg Circuit. The ZPD framework gives us a clear and practical way to understand why these challenges persist and what can be done to address them through better teaching. The next chapter describes how this study will collect and analyse information to answer the research questions.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes how the study will be conducted. It covers the research paradigm, the research design, who will participate, how information will be collected, how it will be analysed, how the quality of the study will be ensured, and what ethical steps will be taken. Every decision made in this chapter is explained and justified with reference to well-known research methodology scholars. Together, these decisions ensure that the study is trustworthy, ethical, and able to produce findings that genuinely address the research questions.

3.2 Research Paradigm and Approach

This study will be conducted within the interpretive research paradigm. The interpretive paradigm is based on the view that people's experiences and the meaning they attach to those experiences are important and worth studying in depth (Creswell, 2014). Rather than measuring a problem with numbers, this paradigm focuses on understanding it from the inside — through the perspectives of the people who live it. This approach is appropriate for this study because what the researcher wants to understand is not a number or a percentage, but the real experiences of learners and teachers as they deal with mapwork every day.

In line with this paradigm, the study will use a qualitative research approach. Qualitative research is suited to questions that begin with 'how' or 'why', and it allows for rich, detailed descriptions of people's experiences and contexts (Merriam, 2009). Since the study aims to understand why learners in the Pietersburg Circuit struggle with mapwork, and how teachers currently approach these challenges, qualitative research is the most appropriate choice.

3.3 Research Design: Interpretive Case Study

The specific design that will be used is an interpretive case study. A case study focuses on a specific, clearly defined situation or group and tries to understand it in as much depth as possible (Creswell, 2014). In an interpretive case study, the goal is not to judge or evaluate, but to understand (Arya, 2021).

In this study, the 'case' will be Grade 10 Geography mapwork learning in selected schools in the Pietersburg Circuit. The case is clearly defined by grade (Grade 10), subject (Geography), location (Pietersburg Circuit), and topic (mapwork). This focus allows the study to produce detailed, meaningful findings that are grounded in the real conditions of a specific local context — something that broader national studies cannot achieve (Merriam, 2009).

3.4 Population and Sampling

The population of this study includes all Grade 10 Geography learners and their teachers at secondary schools in the Pietersburg Circuit, Capricorn South District, Limpopo Province. Purposive sampling will be used to select participants. This means that participants will be chosen because of their specific knowledge, experience, and relevance to the research questions — not randomly (Bryman, 2016).

The sample will consist of three to five Grade 10 Geography teachers from selected schools in the Pietersburg Circuit, who actively teach mapwork and are willing to take part. In addition, learner mapwork test scripts from those schools will be selected for document analysis. Purposive sampling is appropriate here because the goal is to gain deep understanding — not to generalise findings to a large population (Creswell, 2014). The sample size is consistent with accepted norms for qualitative case study research, where depth and richness of data matters more than the number of participants.

3.5 Data Collection Methods

Two data collection tools will be used in this study, each suited to a different part of the research problem.

Instrument	Description	Justification
Semi-structured Interviews	Individual interviews will be held with Grade 10 Geography teachers to find out their views on why learners struggle with mapwork, how they teach it, and what resources they have available.	Allows the researcher to ask flexible, in-depth questions while still staying focused on the research topic. Follow-up questions can be asked when something interesting comes up (Merriam, 2009).

Document Analysis	Grade 10 learner mapwork test scripts and answer books will be examined to find patterns in the types of mistakes learners make across different mapwork topics.	Provides concrete evidence of where learners go wrong without needing to involve learners directly, which reduces ethical complications. Gives an objective picture of learning difficulties (Bryman, 2016).
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The interviews will follow a guide that covers four main areas: which mapwork topics teachers believe are hardest for learners; how teachers currently teach those topics; what resources they have available; and what they have observed about how learners behave and think during mapwork lessons. All interviews will be audio-recorded with the participant's permission and then written out in full for analysis.

The document analysis will look at a selected set of learner test scripts from Grade 10 Geography mapwork assessments at the chosen schools. A structured framework will be used to identify and group the types of errors learners make, based on the four problem areas described in Chapter 2: scale and distance errors, contour interpretation errors, misreading of instructions, and missing steps in procedures.

3.6 Data Analysis

Interview data will be analysed using thematic analysis, as described by Braun and Clarke (2006). This process involves reading through the interview transcripts carefully, identifying patterns and ideas that come up repeatedly, grouping those patterns into themes, and then interpreting what those themes mean in relation to the research questions. Naeem et al. (2023) provide a practical step-by-step guide to this process that will be followed throughout the analysis.

The test script data will be analysed using the error analysis framework described above. The frequency and types of errors will be recorded and used to build a clear picture of the most common and serious difficulties learners face in mapwork.

Throughout the analysis, Vygotsky's ZPD framework will be used as a lens to help interpret the findings — specifically, to understand where the gap is between what learners can do on their own and what they need support to achieve, and what kinds of teaching support are currently missing.

3.7 Ensuring the Quality of the Study

In qualitative research, the quality of a study is measured in terms of trustworthiness (Lincoln & Guba, 1985). This study will use four strategies to ensure trustworthiness. First, credibility will be supported by using two different data sources (interviews and document analysis) and by sharing interview summaries with participants to check that their views have been captured accurately. Second, transferability will be supported by providing detailed descriptions of the research context, so that readers can judge whether the findings might apply to similar settings. Third, dependability will be ensured by keeping a clear record of all methodological decisions. Fourth, confirmability will be supported by the researcher being transparent about their own background and assumptions throughout the study.

3.8 Ethical Considerations

This study will be conducted in line with the ethical requirements for research involving people. Before any data is collected, ethical clearance will be obtained from the University of Limpopo Research Ethics Committee. Permission will also be obtained from the Limpopo Department of Education and the principals of the selected schools. The following ethical measures will apply throughout the study:

Ethical Principle	How it will be addressed
Informed Consent	All teacher participants will sign a written consent form before taking part. Where learner test scripts are used, the school principal's consent will be obtained. Individual learner names will not be recorded.
Voluntary Participation	Taking part will be entirely voluntary. Participants will be told they can withdraw at any time, without any consequences.
Confidentiality and Anonymity	All participants and schools will be given pseudonyms. Data will be stored on a password-protected device that only the researcher can access.
Avoidance of Harm	Interview questions will only focus on professional and teaching experiences. No personal or sensitive questions will be asked.
Honesty and Integrity	All findings will be reported honestly. No data will be made up or misrepresented.

EXPECTED OUTCOMES

It is expected that this study will:

- Identify the specific mapwork topics, such as contour interpretation, scale, cross-sections, and bearings that Grade 10 Geography learners in the Pietersburg Circuit find hardest, supported by evidence from learner test scripts.
- Show how weak mathematical skills and difficulties with examination language add to learners' mapwork challenges, beyond just spatial thinking difficulties.
- Describe the teaching approaches currently used by Grade 10 Geography teachers in the Pietersburg Circuit for mapwork, and identify gaps between what is recommended and what is actually happening in classrooms.
- Show where learners most need additional teaching support in the mapwork curriculum, using the ZPD framework to explain why current approaches are not working.
- Produce a set of practical, realistic recommendations for teachers, curriculum advisors, and the Limpopo Department of Education; recommendations that take into account the real resource constraints of schools in the Pietersburg Circuit.

The findings are expected to contribute to better teaching strategies, more focused curriculum support, and improved learner performance in mapwork across the Pietersburg Circuit and other similar schools in Limpopo Province.

RESEARCH PLAN (TIME SCHEDULE)

Activity	Apr	May - Jun	Jul - Aug	Sept	Oct	Nov
Literature review and proposal refinement	✓	✓				
Develop interview schedule and error analysis framework		✓				
Obtain ethical clearance and school permissions		✓				
Pilot study (testing the instruments)			✓			
Data collection: interviews and document analysis			✓	✓		
Data analysis: thematic and error analysis				✓	✓	
Writing up results and discussion					✓	
Final report submission						✓

BUDGET

Item	Description	Estimated Cost (ZAR)
Stationery and printing	Interview guides, consent forms, error analysis sheets, and final report printing	R 450
Transport	Travel to selected schools in the Pietersburg Circuit for data collection	R 800
Data recording	Voice recorder batteries and USB storage device	R 300
Communication	Airtime and data for contacting participants and schools	R 250
Miscellaneous	Unforeseen expenses	R 200
TOTAL		R 2 000

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